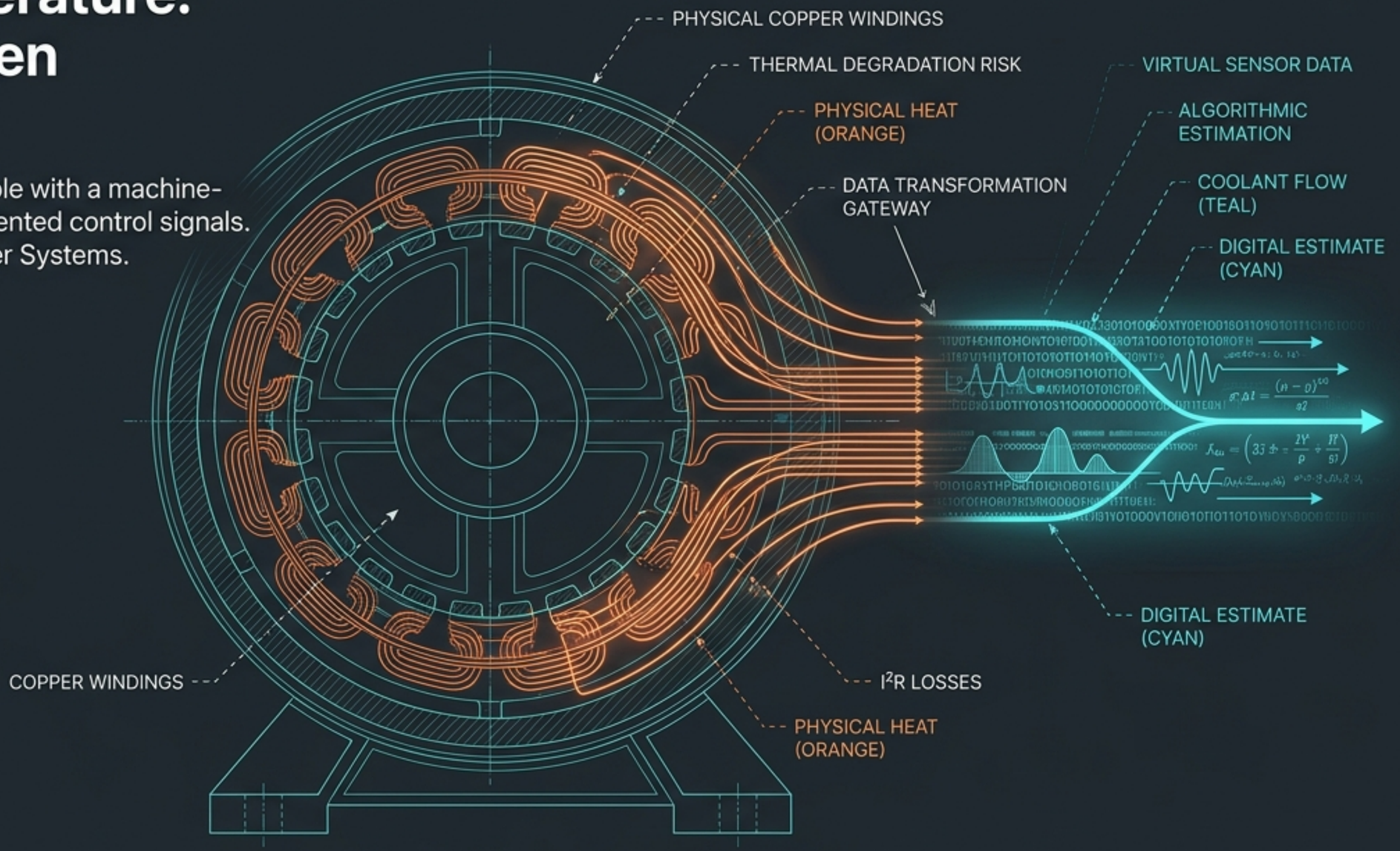
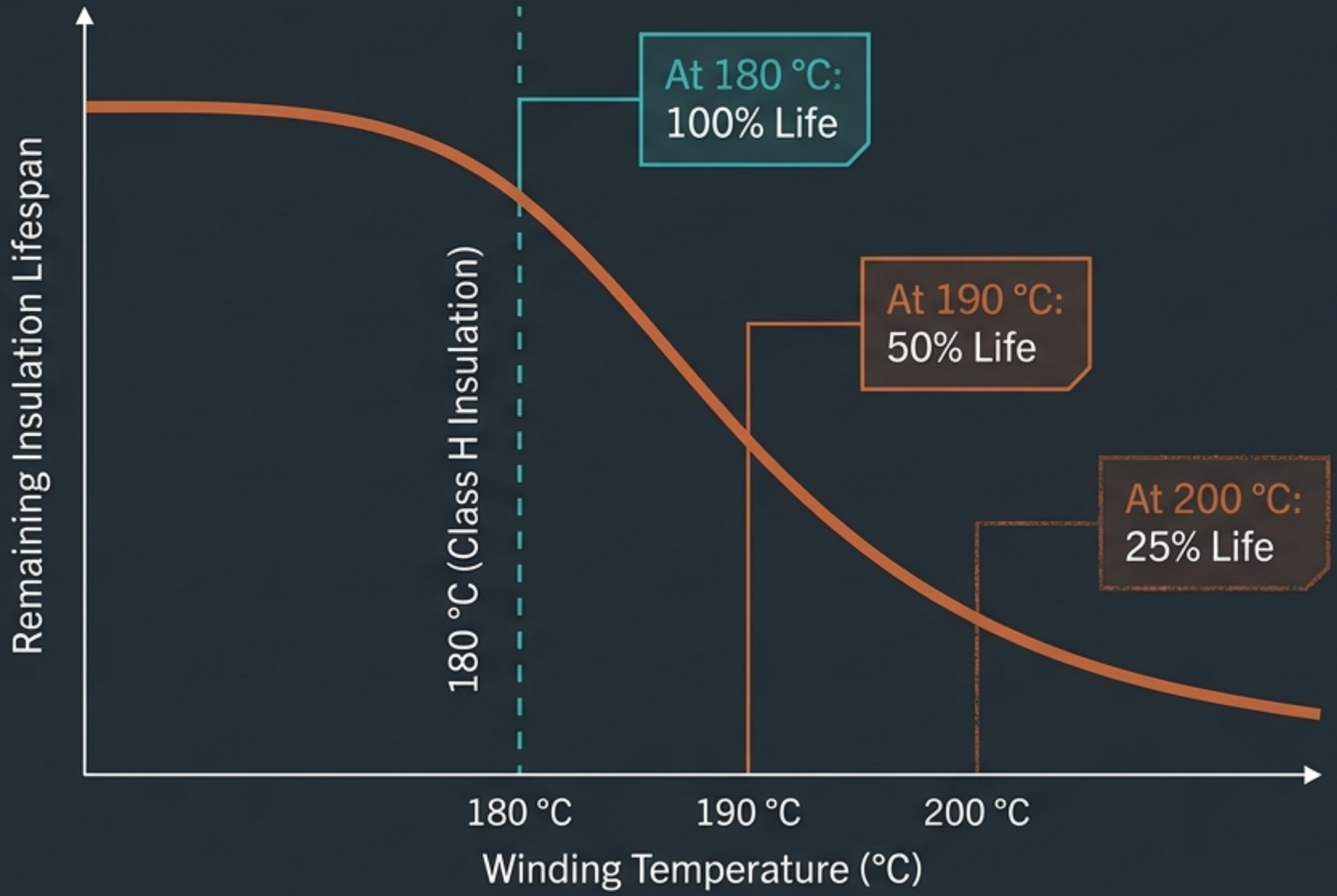
Predicting Electric Motor Winding Temperature: A Physics-Driven Virtual Sensor

Replacing a physical thermocouple with a machine-learning estimator using field-oriented control signals.
Problem Statement 6 — GE Power Systems.



Gemini Notebook

The hardware constraint: insulation life halves for every 10 °C of overtemperature.



Without accurate temperature:

Controllers must apply a fixed, conservative derating, giving up physical torque.

With accurate temperature:

Controllers can push the motor into its short-term overload capability, converting data directly into usable power density.

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Physical sensors survive the test bench, but fail the production bill of materials.

The Hardware Trap



High Voltage Risk

The sensor sits directly on an active electrical conductor carrying high potential.



Harsh Environment

Must endure thousands of extreme thermal cycles (Ambient to 150 °C) and relentless vibration.



Seal Leakage

Routing physical leads through the motor casing creates contamination paths and failure points.

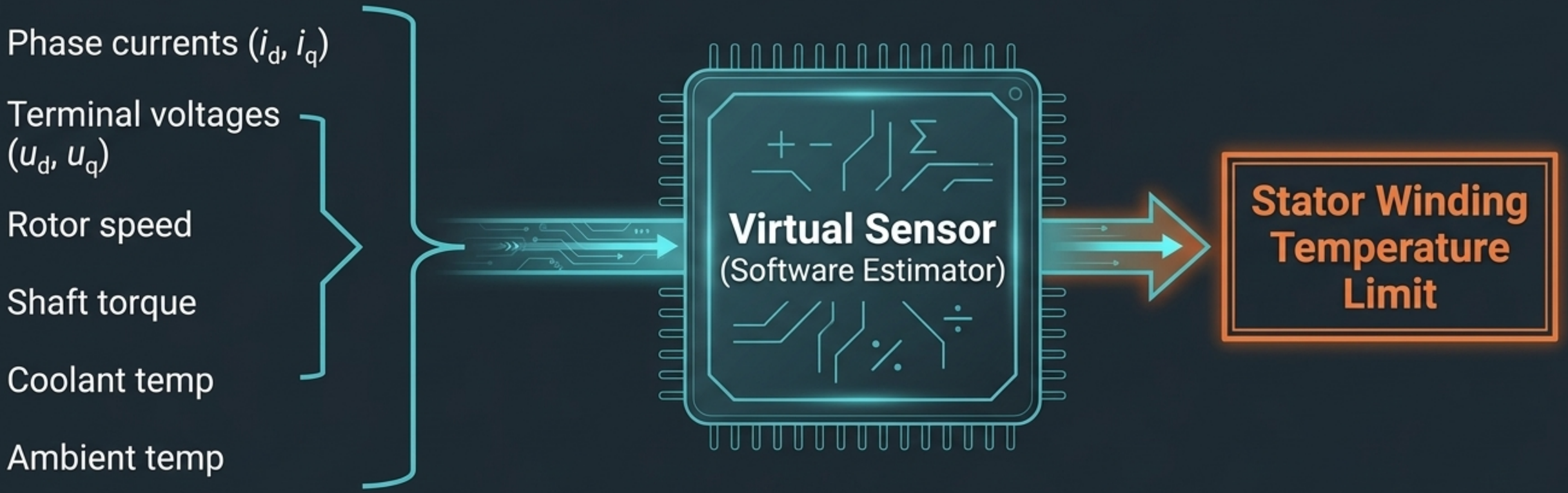


Manufacturing Cost

Added parts and assembly steps destroy the margins required for high-volume manufacturing.

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The Objective: Deriving thermal reality from eight free drive signals.



Zero marginal cost. Every one of these inputs is already measured continuously by the motor drive to execute basic control. The goal is a software replacement.

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Decision 1: Refusing to cheat by discarding target-leaking sensors.



The Trap

These internal thermocouples correlate with the target winding temperature at $r > 0.95$. A naïve model would use them to achieve a near-perfect R-squared instantly.

The Reality

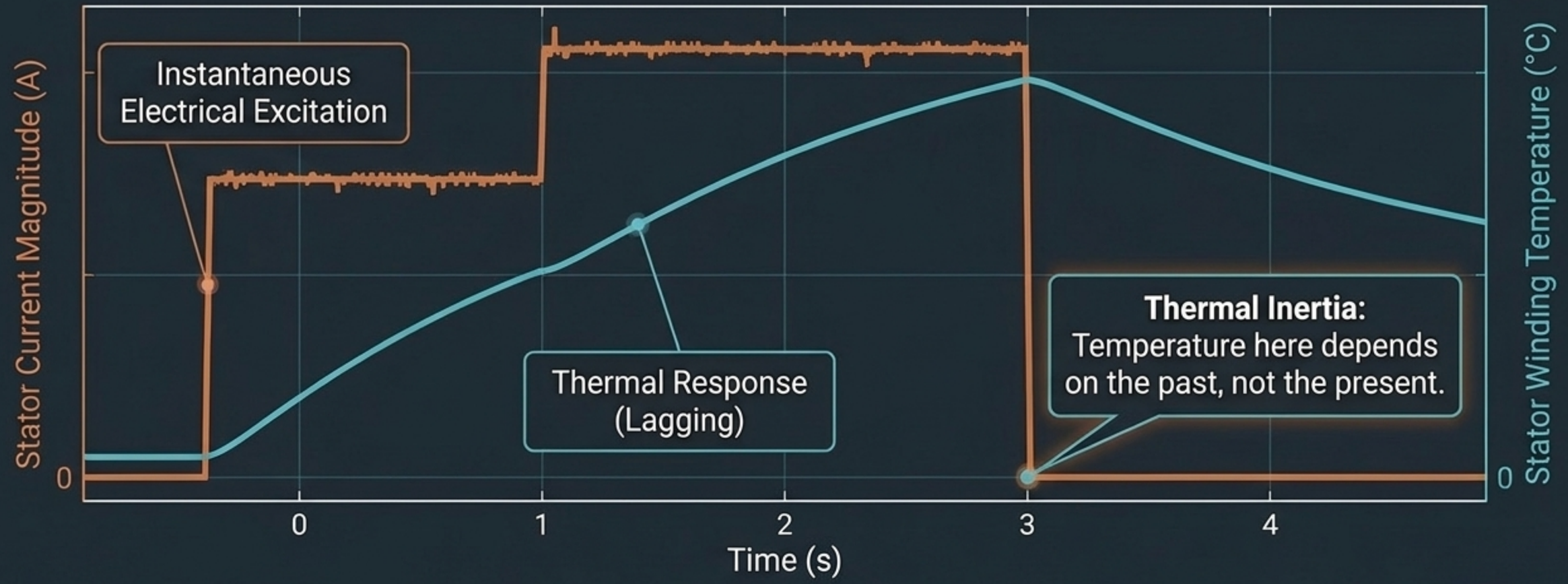
Keeping them defeats the objective (we want to eliminate internal sensors) and constitutes target leakage (they sit millimetres from the winding in the same block of iron).

The Rule

The model must learn the laws of thermodynamics, not just interpolate between adjacent physical thermometers.

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The physical hurdle: current changes in seconds, temperature answers in minutes.



Asking a model to predict current state from instantaneous current is like guessing a bathtub's level by only looking at the running tap.

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Decision 2: Encoding the laws of thermodynamics into the feature space.

Physics

Lumped-parameter differential equation form:

$$C_{th}\frac{d\vartheta}{dt} = P_{loss}(t) - \frac{\vartheta - \vartheta_{amb})}{R_{th}}$$

Calculus

Analytical solution

$$\vartheta(t) = \frac{1}{\tau}\int_{-1}^{\tau} \exp\left(-\frac{t-s}{\tau}\right) P_{loss}(s)ds$$

Code/Math

Discrete software translation

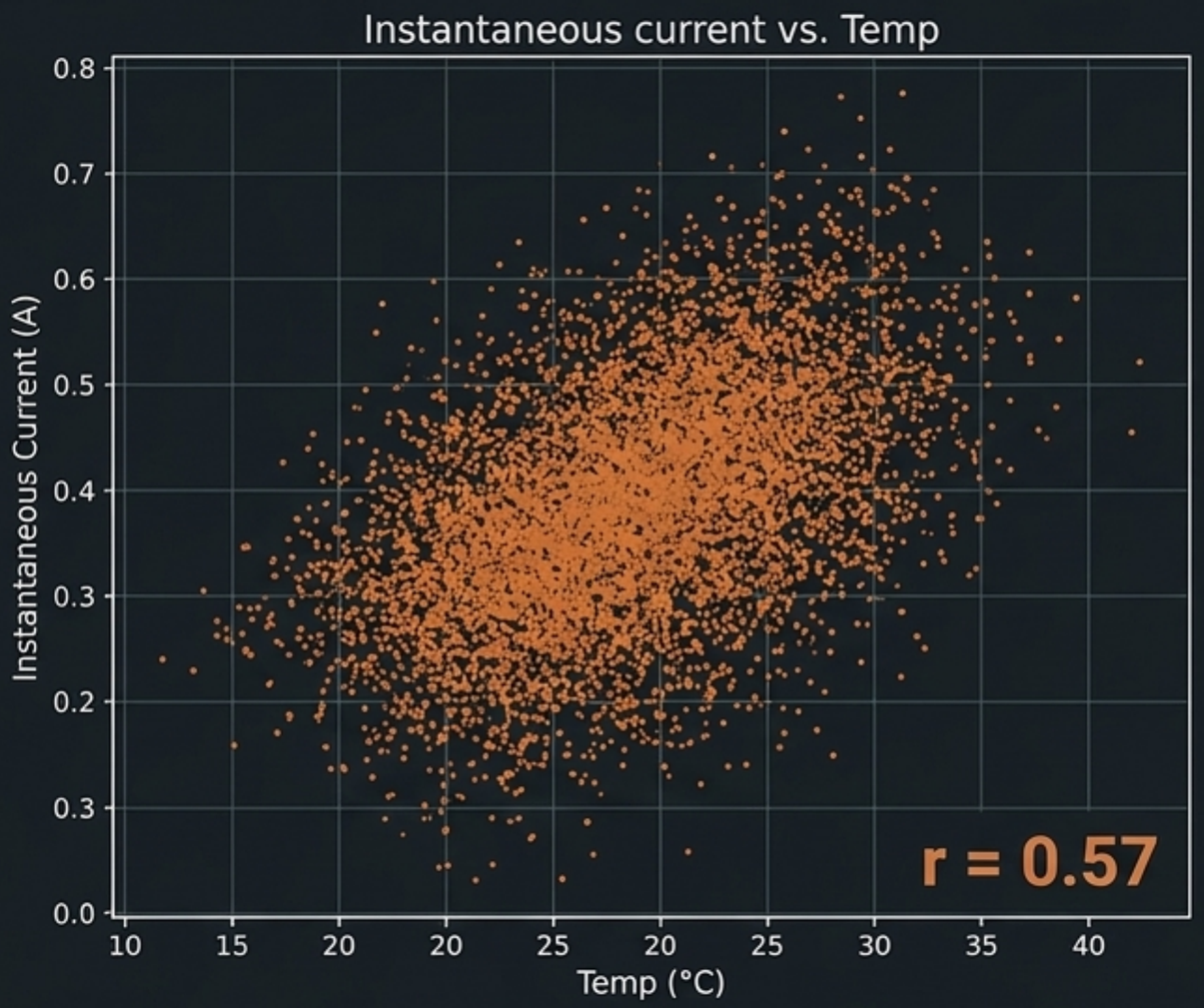
$$y[k] = \alpha \cdot x[k] + (1-\alpha) \cdot y[k-1]$$

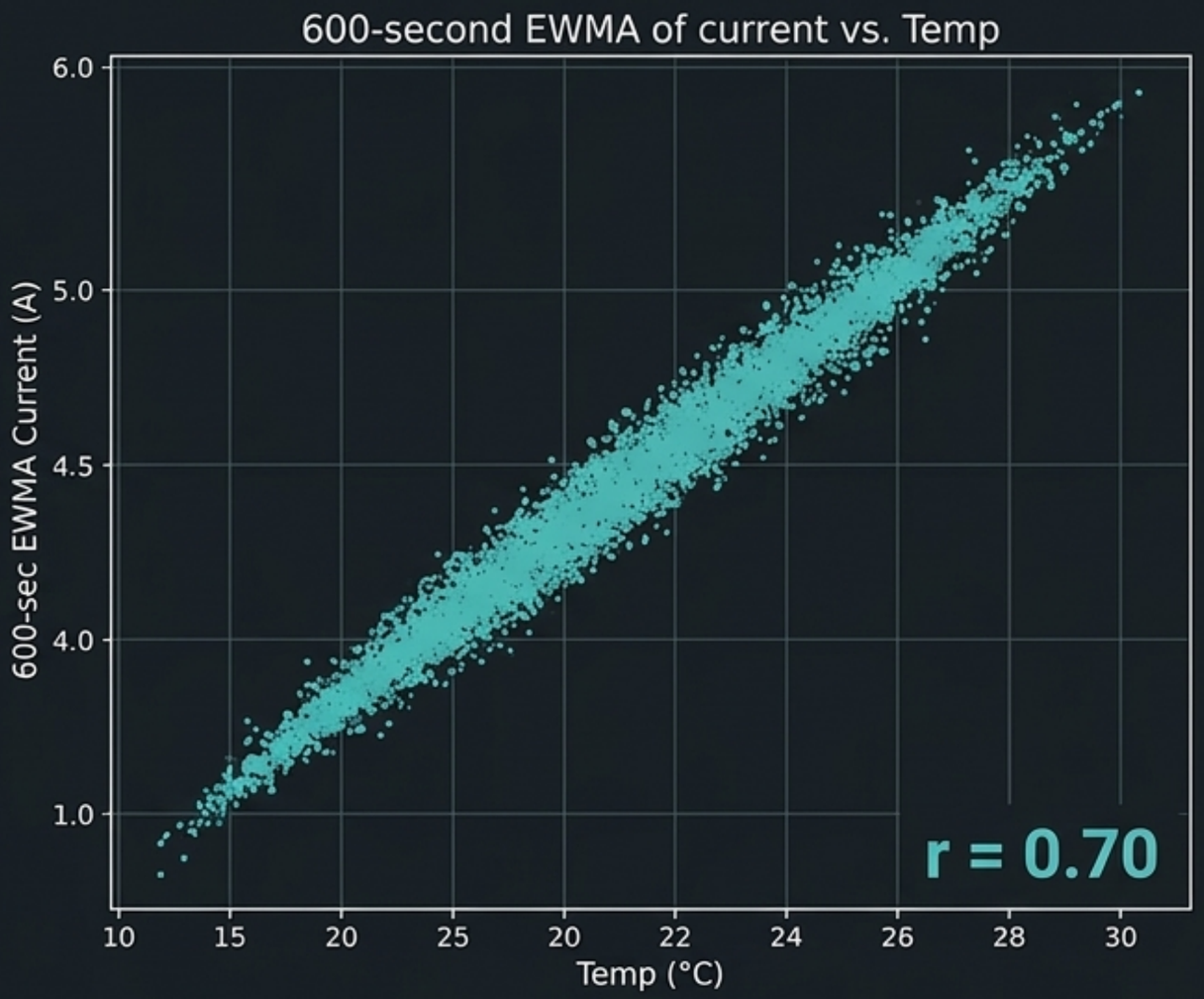
Calculated EWMAs across 5 thermal time spans (30s, 120s, 600s, 1800s, 3600s) to bracket the unknown thermal masses of copper, iron, and coolant.

Expanded 8 raw channels into 104 physics-engineered features.

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The impact of thermal memory on the raw data.



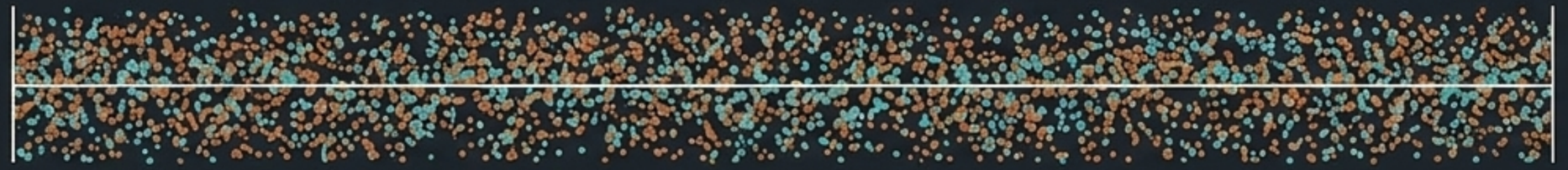


One 10-minute exponential filter transforms a diffuse cloud into a usable relationship. Feature engineering does the heavy lifting.

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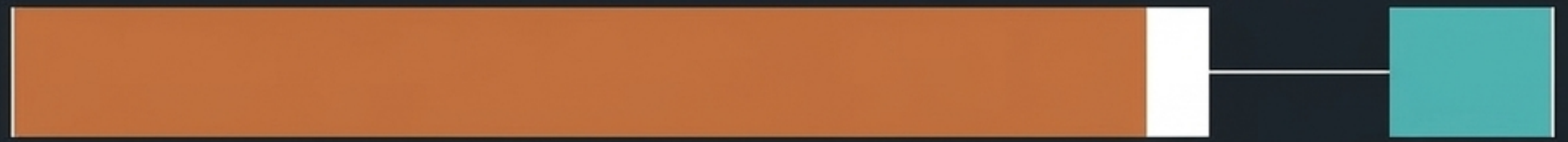
Decision 3: Honest evaluation requires holding out entire motor runs.

Random Split Trap



At 2 Hz, adjacent half-second samples are near-identical twins.
Random splits just measure memorisation.

Profile-Wise Split



64 Training sessions

3 Validation sessions

2 completely unseen Test sessions

Evaluated exactly how it will be used: on motor sessions it has never seen.

Random split (leaky):

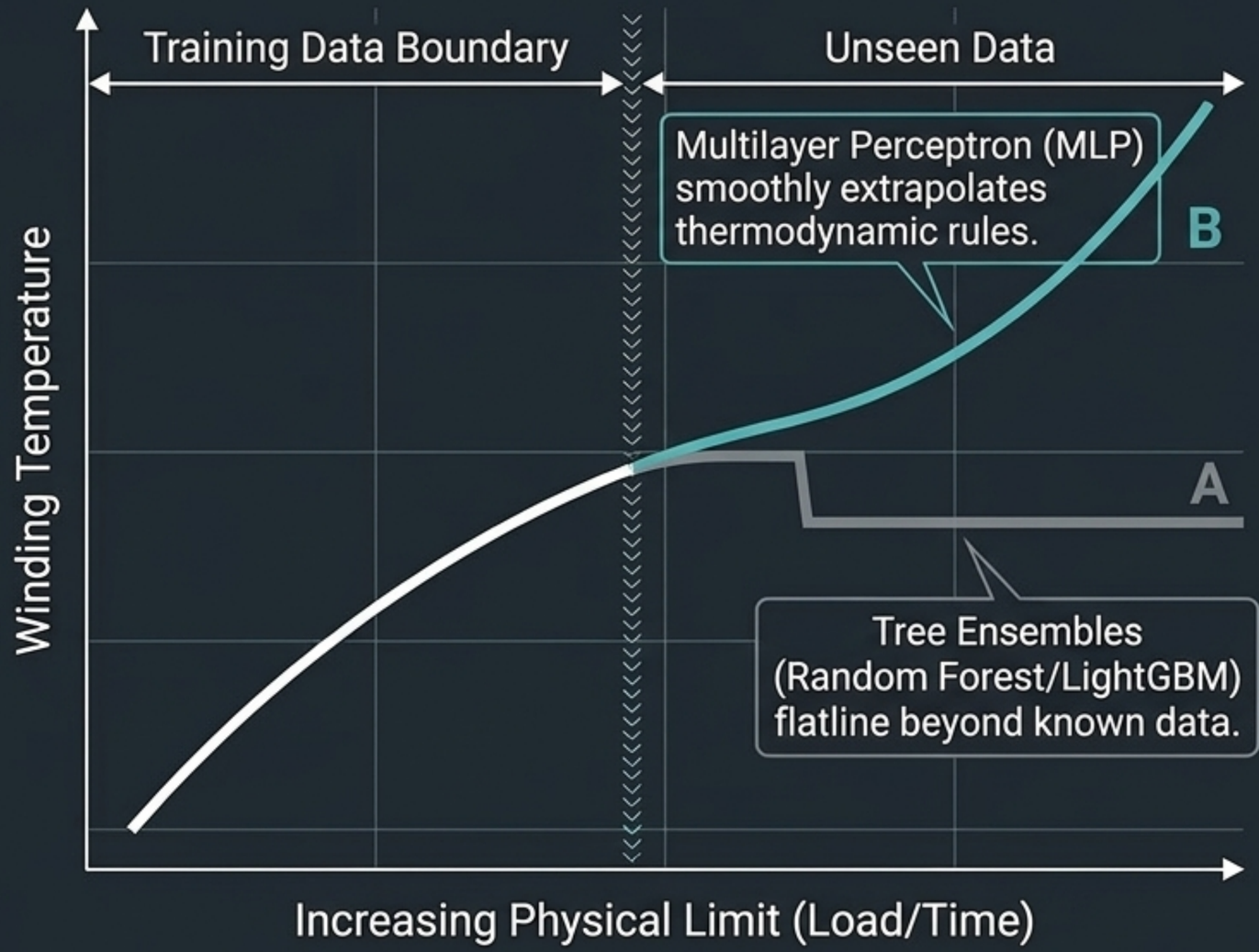
$R^2 = 0.9995$

Profile-wise split (honest):

$R^2 = 0.9908$

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The Extrapolation Problem: Why smooth math beats tree ensembles.



Model Ladder

- Linear Regression on raw inputs: Fails completely ($R^2 = 0.53$).
- Linear Regression with EWMA: Jumps massively ($R^2 = 0.9936$). Feature engineering beat model sophistication.
- The Winner: MLP (2x128). A smooth function approximator that generalises safely into unseen thermodynamic regimes.

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Defining success by physical safety, not just average error.

Naïve ML Approach

- Random Splits
- Instantaneous Features
- Keep Leaky Sensors
- Optimise for **RMSE**

Result: Motor destroys itself on the first unseen load spike

Engineering ML Approach

- Profile Splits
- EWMA Thermal Physics
- Drop Leaky Sensors
- Optimise for **Deployability** and **Worst-Case Error**

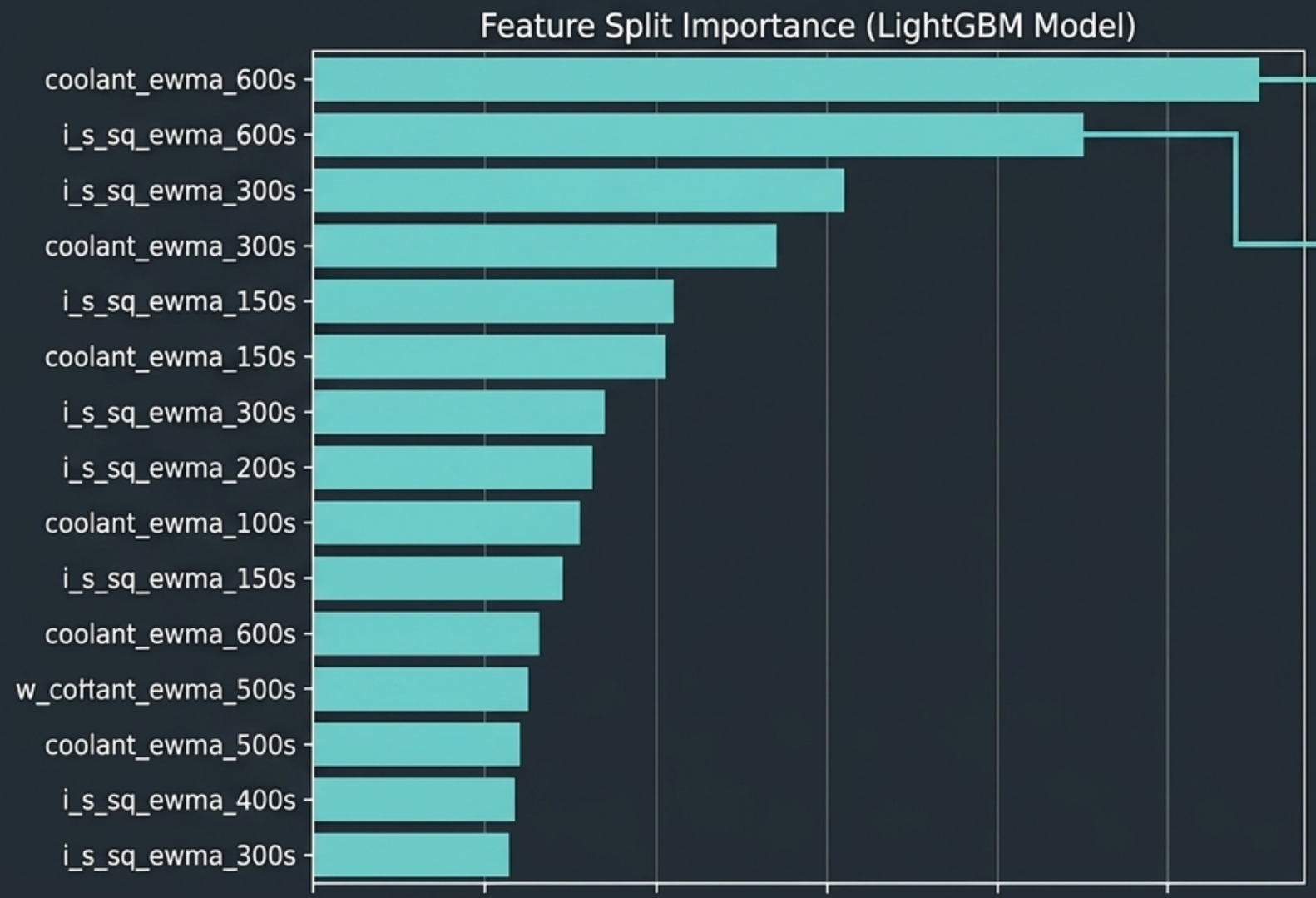
Final Multi-Layer Perceptron (MLP) model

RMSE:	Worst-Case Error:	Deployability:
1.67 °C	**9.84 °C** (The safety margin metric)	**81.3%** of time within ± 2 °C

A 1 °C average error means nothing if a single 20 °C underestimate melts the enamel.

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The algorithm independently rediscovered the laws of thermodynamics.



The heat-sink boundary condition (Heat escaping).

Current squared: $P = I^2R$ (Copper heat generation).

The data-driven model, given no explicit thermodynamic rules, converged on the **exact two driving terms** from the lumped-parameter physics equation.

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Deployment Reality: Why this actually fits on a low-cost microcontroller.

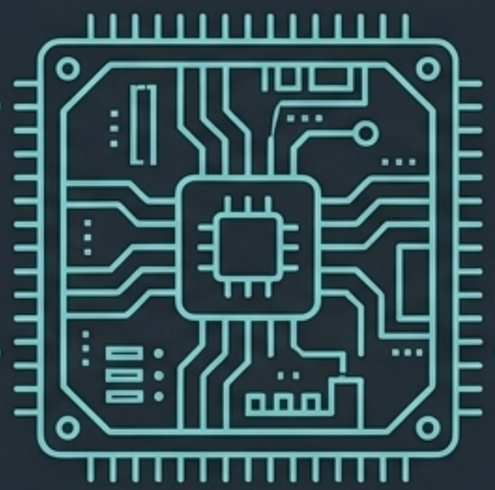
Memory footprint

Just 90 state registers required for the whole 104-feature extractor. No massive sliding-window buffers needed.

Processing load

~60,000 MACs/sec (Multiply-accumulates).

A trivial fraction of modern embedded capacity.



Algorithmic efficiency

Because EWMAs are recursive Infinite Impulse Response (IIR) filters, they require only 1 state value per feature.

Data format

16-bit fixed point (Q16.16 accumulators).

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Executive Summary & Honest Limitations

Success

- Eliminated physical sensors using 8 existing drive signals.
- Physics-derived thermal memory improved RMSE by 6x.
- Proven extrapolation onto entirely unseen motor sessions.
- Microcontroller-ready computational footprint.

Honest Engineering Limits

- **Safety Margin:** 9.84 °C worst-case error requires retaining a safety derating, albeit a much smaller one.
- **Warm-up Lag:** Highest errors occur in the first minutes while 3600s EWMA filters initialise from zero.
- **Single Architecture:** Trained on one physical machine; adapting to new motor geometries requires transfer learning.

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